



General Use SOP

Highly Reactive & Unstable Materials

Process or Experiment Description

This standard operating procedure (SOP) provides general guidance on working safely with lab-scale quantities of highly reactive/unstable materials. In some cases, multiple SOPs may apply to a single chemical (e.g., for perchloric acid, both General Use SOPs for highly reactive/unstable materials and corrosives would apply). For questions about the applicability of any item in this SOP, contact your Principal Investigator/Supervisor or EH&S at 650-723-0448.

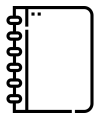
For work with larger quantities of highly reactive/unstable materials, complex procedures, or otherwise particularly hazardous procedures involving highly reactive/unstable materials, EH&S recommends completing a [risk assessment](#), reviewing other EH&S guidance (such as [guidance on scale-up](#)), looking for SOPs on similar procedures in the [SOP Library](#), and/or writing a [dedicated SOP](#) and [submitting it for EH&S review](#).

Hazards

Highly reactive and unstable materials have the potential to vigorously polymerize, decompose, condense, or become self-reactive under conditions of physical impact, pressure, temperature, light, or contact with another material. These reactions pose a higher-than-average risk of running out of control or releasing large amounts of heat, gases (including flammable and/or toxic gases), etc.

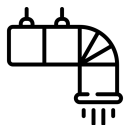
Examples of such substances are explosives (e.g., [picric acid](#)), strong oxidizers (e.g., $\geq 70\%$ nitric acid, $\geq 30\%$ hydrogen peroxide, bleach), peroxides (e.g., hydrogen peroxide, peracetic acid), water-reactives (e.g., metallic sodium, acetic anhydride), self-reactives (e.g., acrolein), and pyrophorics (e.g., lithium aluminum hydride). Some materials that are normally stable can become unstable as they age, such as [peroxide formers](#) (e.g., diethyl ether, tetrahydrofuran). Additionally, certain chemical functional groups are often unstable (e.g., [azide](#), azo, nitro, nitroso, haloamine, peroxides, cyclopropyl).

Control of Hazards



General

- Minimize the potential for splash, splatter, or other likely scenarios for accidental contact.
- Use less reactive alternatives if feasible.
- Work with highly reactive/unstable materials only when there is someone else in the lab ready to assist you in the event of an incident.
- Use the smallest quantity and concentration feasible .
- Handle reactive chemicals with caution, taking appropriate chemical-specific precautions when mixing even small quantities with other chemicals.
- Conduct reactions that require non-ambient temperatures or pressures in a manner that minimizes risk of explosion, implosion, or vigorous reaction.
- Do not place highly reactive/unstable materials inside sealed vessels. If sealing is necessary, a means of pressure relief is necessary and you should consult with EH&S before proceeding.
- Provide a mechanism for adequate temperature control and heat dissipation.
- Use shields and barricades when there is a risk of explosion or vigorous chemical reaction. See the [EHS Safety Store](#) for a loaner [blast shield](#).
- Shield glass equipment that is used under vacuum or elevated pressure; wrap with tape or otherwise protect it from shattering.
- When quenching a highly reactive/unstable material as part of an experimental operation, do so slowly and with adequate temperature control and heat dissipation.
- Materials that are in the original manufacturer's container do NOT need to be quenched before disposal - this process often leads to fires. If you are not sure how to safely package a material for disposal, contact EH&S.
- Have an appropriate fire extinguisher readily available when working with pyrophoric materials (e.g., a Class D extinguisher for metallic lithium, sodium, potassium or other flammable metal).



Engineering/Ventilation Controls

Many reactive materials liberate flammable and/or toxic gas when exposed to water vapor or air. Use a properly functioning [chemical fume hood](#) to prevent hazardous buildup of gases. If the process doesn't

allow working in a fume hood, contact Environmental Health and Safety at 650-723-0448.



Personal Protective Equipment

In addition to proper street clothing (long pants or equivalent that covers legs and ankles, and close-toed non-perforated shoes that completely cover the feet), wear the following Personal Protective Equipment (PPE):

- Safety goggles and face shield
- Flame-resistant lab coat (such as Nomex)
- Appropriate [chemical-resistant gloves](#)

Additional PPE for pyrophoric and/or water-reactive substances:

- Fire resistant gloves may be necessary depending on the task
- Non-synthetic street clothing

Additional PPE for explosive substances:

- Heavyweight gloves (such as anti-static PVC gauntlets)
- Blast shield



Special Handling Procedures and Storage Requirements

Carefully handle materials that may be sensitive to shock, heat, friction, or light. Ensure that materials are chemically compatible with the equipment and tools used in the experiment.

Containers should be closed when not in use. [Segregate](#) incompatible chemicals and use [secondary containment](#). Follow any substance-specific storage guidance provided in SDS.

Label all chemicals with date received and date opened. For peroxide-forming chemicals, assign an appropriate expiration date per manufacturer instructions or using the [SU Safety Fact Sheet on Peroxide Forming Chemicals](#). Do not use or disturb any peroxide-forming chemicals that show crystallization, visible discoloration, or liquid stratification, as these may indicate peroxidation.

Emergency Procedures

All incidents should be reported to EH&S with an [incident report form](#).

Spill

Refer to EH&S' guidance on [hazardous materials incidents](#) for general procedures for incident response, and call 650-725-9999 for assistance with spill cleanup.

If you are trained in spill cleanup and the spill is not health-threatening, proceed with spill cleanup. There may be variations in procedure based on the specific spill, but the general steps are:

1. Don the appropriate PPE (See Personal Protective Equipment section, above).
2. Locate the nearest spill kit and bring it to the spill.
3. Use the absorbent pads in the spill kit to wipe up the spill.
4. Place the used absorbent pad and gloves inside the spill kit jar.
5. Place a waste tag on the spill kit and request waste pickup.

Exposure

If immediate medical attention is required, call 911. Remove any contaminated clothing, and IMMEDIATELY flush contaminated skin with water for at least 15 minutes. Use a safety shower for any skin exposures to the trunk, legs, or feet.

For eye exposures, IMMEDIATELY flush eyes with water for at least 15 minutes using an eyewash station.

If medical attention is needed, provide the SDS(s) of the chemical(s) to aid medical staff in proper diagnosis and treatment.

For minor exposures, personnel can contact the [Stanford University Occupational Health Center](#) at 650-725-5308. For serious exposures, personnel should go to the Stanford Hospital Emergency Department.

Those who work with hazardous chemicals are entitled to receive medical attention/consultation when:

- A spill, leak, explosion or other occurrence results in a hazardous exposure (potential overexposure).
- Symptoms or signs of exposure to a hazardous chemical develop.

Fire

If flammable/combustible liquids catch on fire, it may be extinguished with a CO2 extinguisher if the fire is

small (smaller than a trash can) and you've been trained. Remember PASS:

Pull the pin

Aim at the base of the fire

Squeeze the nozzle

Sweep back and forth

Use the extinguisher from 6 or more feet away. Using the extinguisher too close to the fire risks spraying the ignited liquid.

Waste Disposal

Dispose of highly reactive/unstable materials as hazardous waste. All hazardous waste containers should have a waste tag attached before any waste is added. Before combining materials in waste containers, review the SDSs and Stanford's [Incompatibility Guide](#). For general guidance regarding waste disposal, refer to the guide for [handling and storing waste](#) and the [chemical waste poster](#).

Highly reactive or unstable materials should have their own dedicated waste container, and no other materials should be added unless the researcher is certain there will be no reaction.

Minimum Training Requirements

- General Safety & Emergency Preparedness (EHS-4200)
- Chemical Safety for Laboratories (EHS-1900)
- Laboratory-specific training

Approval Required

Consult with your PI regarding the need for prior approval. Laboratory personnel shall seek and the PI must provide prior approval of any chemical usage involving [Restricted Chemicals or High Risk Procedures](#). EH&S recommends obtaining prior PI approval for work with highly reactive/unstable materials.

Designated Area

For highly reactive/unstable materials that are also considered [particularly hazardous substances](#), a designated area shall be established per other applicable SOPs.